Quantum-Assisted Multi-Asset Portfolio Construction

Step 05Q

Hybrid QAOA Active-Set Selection

Using QAOA to choose a sparse trading sleeve and classical optimization to set weights

Project technical report

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1 The hybrid question

Step 5Q does not ask a quantum circuit to solve the entire 50-asset allocation. The continuous portfolio is already a convex problem with strong classical solvers. The difficult discrete question is which small set of coordinates should be allowed to rebalance. QAOA is applied to that support-selection problem.

2 Candidate screening

Each asset receives an active-selection score based on normalized growth, income, standalone variance, scenario RMS loss, implementation burden, incumbent weight, and marginal utility from small transfer experiments. The screen retains buy and sell opportunities rather than simply choosing the highest-return assets.

The final reduced universe contains 11 qubits. A qubit corresponds to a candidate permission, not a final weight. The selected cardinality is six.

3 Executed active-rebalance binary energy

Final QAOA ranks the 11 screened active-coordinate permissions using the implemented marginal-utility surrogate. The richer diagnostic QUBO is not the headline objective.

$$\begin{aligned} q_i &= -\widetilde{s}_i^{\,2} + 0.05\,\widetilde{c}_i\widetilde{s}_i, \\ Q &= \lambda_b\widetilde{t}\widetilde{t}^{\mathsf{T}} + 0.04(\widetilde{t}\widetilde{t}^{\mathsf{T}})\odot\widetilde{\Sigma} + 0.02(\widetilde{t}\widetilde{t}^{\mathsf{T}})\odot\widetilde{\Gamma}. \end{aligned} \tag{1}$$

Active-balance strength = 1.0 and QAOA aggregation = 0.25. Exact trade sizes, scenario hinges, and all Step 1 constraints are restored by continuous refinement.

4 Circuit configuration

The final evidence run used:

Qubits	11
Selected cardinality	6
QAOA depth p	2
Shots per run	1,024
Classical optimizer iterations	60
Independent seeds	20
Feasible supports	462

The Dicke initial state and XY-ring mixer preserve cardinality in the ideal circuit. Qiskit Aer statevector-based sampling was used; the experiment did not run on quantum hardware.

5 Classical optimization inside QAOA

QAOA remains a hybrid algorithm. For each seed, a classical optimizer updates the $2p$ angles. The circuit estimates an energy distribution; the classical loop proposes new angles. After sampling, the best bitstrings are decoded and passed to the continuous optimizer.

6 Exact enumeration and baseline selectors

All 462 six-of-eleven supports are enumerated exactly for the executed reduced-QUBO energy E_QUBO(z). Greedy and local search recover the same QUBO-minimizing support. Only retained QAOA and high-ranked supports are sent to continuous refinement. This certifies reduced-QUBO optimality, not global optimality of the exact refined financial objective across all 462 supports.

7 Reliability across seeds

Fifteen of 20 seeds reached the exact minimum energy, an exact-hit rate of 75%. The remaining seeds found second- or lower-ranked feasible supports. Reporting the seed distribution is more informative than reporting only the best run.

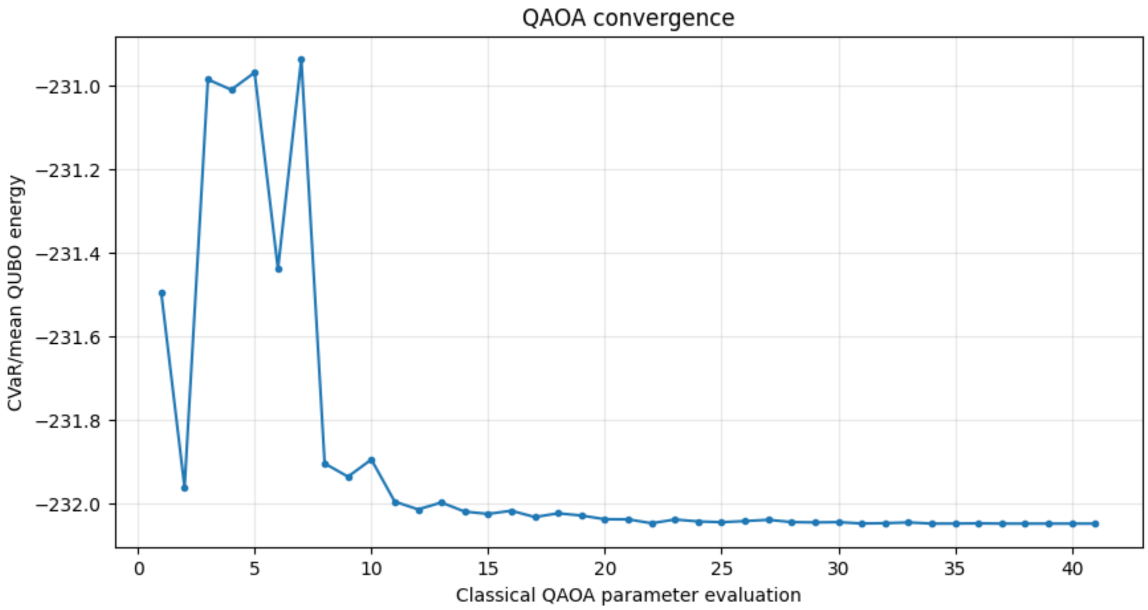

Figure 1: Seed-level QAOA energy outcomes relative to exact enumeration.

A best-run exact result does not mean the optimum had a dominant single-shot probability. In the retained sample distribution, the exact state can appear only once in 1,024 shots. The 75% seed-level hit rate concerns whether the optimization-and-sampling process produced the optimum at least once, not whether the final state was highly concentrated on it.

8 Selected support and material trades

The selected support is

$$\{\mathrm{SGOV}, \mathrm{BIL}, \mathrm{SHY}, \mathrm{QQQ}, \mathrm{VUG}, \mathrm{MTUM}\}. \quad (2)$$

The exact continuous refinement produces four material trades:

Ticker	Current	Hybrid	Trade	Absolute	Selected	Executed
SGOV	1.43%	6.96%	+5.53%	5.53%	True	True
VUG	2.37%	0.00%	-2.37%	2.37%	True	True
QQQ	2.06%	0.00%	-2.06%	2.06%	True	True
MTUM	2.26%	1.16%	-1.10%	1.10%	True	True
SHY	3.40%	3.40%	-0.00%	0.00%	True	False
BIL	1.57%	1.57%	-0.00%	0.00%	True	False

BIL and SHY remain selected permissions but receive zero material incremental trade. This is not contradictory: support selection defines the feasible coordinate set, and the continuous optimum may sit at the incumbent value on some permitted coordinates.

9 Portfolio result and opportunity cost

Portfolio	Return	Volatility	Worst stress	Turnover	Cost	Effective holdings
Primary classical	5.49%	7.86%	12.71%	13.09%	0.0020%	33.73
Hybrid QAOA	5.45%	7.87%	12.72%	11.06%	0.0015%	35.48
Strict warning	5.48%	7.51%	12.17%	23.88%	0.0056%	29.35

Relative to the unrestricted primary classical portfolio, the hybrid loses roughly 4 basis points of expected return, adds about 1 basis point of volatility and stress loss, but reduces turnover by about 2.03 percentage points and cost by about 0.0005 percentage point. The normalized objective gap is approximately 0.000970. This is the cost of restricting the active sleeve.

10 Constraint interpretation

All hard constraints pass. The portfolio has five soft warning breaches, with substantial hard-limit headroom. The precise wording is: *hard-limit compliant under the declared soft-warning policy, with five soft-warning breaches.*

11 Why the result is technically meaningful

The quantum layer influenced the actual feasible set used by the continuous optimizer. It was not added after the portfolio was solved. The end-to-end flow is

$$\text{50 assets} \rightarrow \text{11 candidates} \rightarrow \text{QAOA selects 6} \rightarrow \text{continuous refinement} \rightarrow \text{full audit.} \tag{3}$$

12 Why the result is not quantum advantage

The binary search space has only 462 states, which are exactly enumerable. Aer is a simulator, not hardware. The project therefore demonstrates a correct hybrid decomposition and a reliable selected run, not computational speedup or scale superiority.

13 Reproduction checklist

Preserve candidate ordering, QUBO coefficients, fixed-cardinality convention, initial state, mixer, QAOA depth, shots, optimizer, seed list, and exact enumeration. Save every seed's best energy, support, and continuous refinement. Confirm the best selected energy equals the exact minimum and rerun the full hard-constraint audit.

A Why the quantum problem is an active-rebalancing problem

A direct 50-asset weight encoding would require discretizing continuous weights, adding many qubits, and embedding budget, turnover, class, liquidity, and stress constraints in a large penalty model. That is not credible for the available simulator. The project instead keeps the full 50-asset portfolio and asks which of 11 candidate coordinates may change. Inactive assets remain at incumbent weights. This makes the binary question economically meaningful and computationally testable.

B Candidate screening

Candidate trades are ranked using finance-aware information such as marginal objective improvement, growth and income contribution, risk and scenario effect, class balance, implementation cost, and pairwise diversification. Screening is not the final optimizer; it reduces the binary universe. A reproducible screen must save all 11 candidates and their component scores so the reduction is auditable.

C Cardinality and feasible state count

The binary vector satisfies

$$z \in \{0,1\}^{11}, \qquad \mathbf{1}^{\mathsf{T}} z = 6. \tag{4}$$

The feasible state count is

$$\binom{11}{6} = 462. \tag{5}$$

The top 11 assets by incumbent mass could not simply be selected and liquidated or replaced as a whole because their current mass would imply turnover above the limit. The active-permission interpretation avoids this problem: selection allows a coordinate to move, while the continuous optimizer controls the amount.

D QAOA configuration

The final experiment uses depth $p = 2$, 1,024 shots, a maximum of 60 classical optimizer iterations, and 20 independent random seeds. It starts in a feasible fixed-cardinality state and uses an XY-ring mixer. The backend is a simulator, not quantum hardware. Saving only the best seed would hide reliability; therefore the project stores the complete seed-level table.

E Measurement, decoding, and reranking

For a measured bitstring z, sampled QUBO energy is not sufficient. The workflow is:

1. reject or postselect any string with the wrong Hamming weight;
2. decode the six selected ticker coordinates;
3. solve the exact support-constrained continuous portfolio problem;
4. recompute exact economic objective and every hard constraint;
5. record executed trades separately from selected permissions; and
6. rank valid supports using the exact refined objective.

This explains why six coordinates can be selected while only four have material trades. The optimizer found zero incremental trade to be optimal for BIL and SHY, even though they remained permitted coordinates.

F Final support and portfolio economics

The selected support is

$$\{\mathrm{SGOV}, \mathrm{BIL}, \mathrm{SHY}, \mathrm{QQQ}, \mathrm{VUG}, \mathrm{MTUM}\}. \tag{6}$$

Material changes are approximately: buy SGOV by 5.5276 percentage points; sell MTUM by 1.0996 points; sell QQQ by 2.0565 points; and sell VUG by 2.3715 points. These trades reduce growth-equity exposure and increase cash-like exposure.

The refined QAOA portfolio has approximately 5.45% expected return, 7.87% volatility, 12.72% worst scenario loss, 11.06% gross turnover, and 0.0015% modeled one-time trading cost. It passes every hard constraint but exceeds five soft warning thresholds.

G Seed-level reliability

Let E^* be exact minimum QUBO energy and E_s the best energy sampled in seed s. Define gap

$$\Delta_s = E_s - E^* \geq 0. \tag{7}$$

A seed is an exact hit when Δ_s is within numerical tolerance. Fifteen of 20 seeds are exact hits, giving

$$\widehat{p}_{\mathrm{hit}} = \frac{15}{20} = 75\%. \tag{8}$$

The selected best run has gap zero. This establishes repeatable access to the optimum in this experiment, not deterministic convergence and not a scaling advantage.

H Probability interpretation caution

The fact that an optimum appears in many independently optimized seeds is useful. However, the best-run sample probability of one bitstring should be compared with a matched baseline before claiming concentration. A uniform distribution over 462 feasible states assigns probability about

$$\frac{1}{462} \approx 0.216\% \tag{9}$$

per state. Shot noise, optimizer selection, and best-of-20 reporting all affect observed frequency. The project therefore emphasizes exact-hit rate and energy rank rather than claiming a quantum sampling advantage.

I Classical baselines

Exact enumeration evaluates all 462 reduced-QUBO energies. Greedy and local-search baselines recover the same QUBO-minimizing support as the selected QAOA run. Because these selectors therefore refine to the same fixed support, they yield the same continuous weight vector; Step 6 later deduplicates that portfolio.

J No quantum-advantage claim

Classical enumeration is trivial at 462 states. The quantum value demonstrated here is architectural: a binary active-set role is defined, implemented, benchmarked, and integrated with exact continuous finance constraints. A future advantage study would require larger instances, fair runtime accounting, hardware noise analysis, and classical algorithms appropriate to that scale.

K Reproduction protocol

1. Freeze the Step 5 context and 11 candidate coordinates.
2. Export candidate and pairwise QUBO coefficients and verify energy conventions.
3. Enumerate all 462 feasible states and store E^*.
4. Build the Dicke initial state, XY mixer, and depth-two QAOA circuit.
5. Run 20 declared seeds with 1,024 shots and the same optimizer budget.
6. Decode, refine, and audit every retained support.
7. Report every seed, the exact-hit count, selected energy gap, support, final weights, material trades, and warning/hard status.

L Reduced-QUBO robustness diagnostics

$$\begin{aligned} &\lambda_b \in \{0.5, 1, 2\}, && \alpha_\Sigma \in \{0.02, 0.04, 0.08\}, \\ &\alpha_\Gamma \in \{0.01, 0.02, 0.04\}, && \alpha_c \in \{0.025, 0.05, 0.10\}. \end{aligned} \tag{10}$$

Across all 81 declared coefficient combinations, the submitted six-of-eleven support remains the exact reduced-QUBO ground state. Exact reduced-QUBO minima are also exported for cardinalities 4 through 8. These are surrogate-QUBO robustness diagnostics; they do not replace exhaustive continuous financial refinement of every support.